

Syllabus for Math 4200:
Introduction to Complex Variables
Fall 2025

Instructor: David Schwein (he/him)

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Office: JWB 305

Class time: MWF 11:50–12:40

Class location: LCB 215

Office hours: M 1:30–2:30pm, W 2:00–3:00pm, F 10:30–11:30

In this course we will develop ideas from real analysis using the complex numbers instead of the real numbers. In this alternative number system, many miraculous properties hold: for instance, as soon as a function is differentiable, it has a power series expansion and derivatives of all orders. The key to proving these properties is an operation called *contour integration*, which will be the focus of much of the course.

Prerequisites: A grade of C or better in Math 3220. If you don't meet this prerequisite but would like to take the class, come talk to me.

Course webpage: On Canvas. I will post all course materials there. Make sure that you regularly check for announcements or turn on the relevant notifications.

Format: People learn best by doing, not watching. Although this course is primarily lecture based, questions are warmly encouraged, and I will integrate exercises on current and previous topics into each class meeting. I prefer that you work on these exercises in small groups: Collaborating helps solidify your knowledge and can also be quite fun!

Textbook: Complex analysis is an old subject and there is no practical limit to the number of resources for studying it. I will loosely follow the book

Complex Variables by Joseph L. Taylor (ISBN 978-0-8218-6901-7).¹

Some resources that helped me learn this myself are the textbooks of Ahlfors, Brown and Churchill, Busam and Freitag, and Gamelin, as well as the Schaum's outline series. Wikipedia discusses (accurately) most of the key theorems and techniques from this course.

Homework: Homework will be released each Friday and due at the start of class the following Friday. Your work can be typed or handwritten, but please write in complete sentences: Writing quality will contribute to your grade.

Collaboration policy: To help learn the course material, you are allowed to consult with your peers and professors, additional textbooks, online resources such as Math Stack Exchange or Wikipedia, and AI assistants like ChatGPT. I particularly encourage working with your classmates. Nonetheless, all work that you submit must be your own.

¹An earlier version of the textbook very similar to the published one is freely available online at <https://web.archive.org/web/20100614000721/http://www.math.utah.edu/~taylor/complexanalysis>.

Assessments: In addition to homework, we will have three quizzes, two midterm exams, and one final exam. The quizzes and midterm exams will take place in class on Fridays, on the following dates: quizzes on September 5, October 3, and November 14, and midterms on September 19 and October 31.

The final exam will take place 10:30–12:30 on Monday, December 8, in our usual classroom.

Grade: $(25\% \text{ homework}) + (5\% \times 3 \text{ quizzes}) + (15\% \times 2 \text{ midterms}) + (30\% \text{ final}) = 100\%$.

Course outline: I plan to cover most of the first six chapters of Taylor, excluding sections 5.4, 6.5, and 6.6. However, we will not work through the book in order, and I will cover the topics according to my own preferences.

Introduction	(August 18)	Cauchy's Theorem	(7 lectures)
The complex numbers	(4 lectures)	Power series	(7 lectures)
Topology of $\mathbb{C}$	(3 lectures)	Midterm 2	(October 31)
Holomorphic functions	(2 lectures)	Residues	(7 lectures)
Contour integration	(3 lectures)	Conformal mapping	(6 lectures)
Midterm 1	(September 19)	Final exam	(December 8)

Climate: Each of you deserves to learn in an environment where you are respected and feel safe. As the next generation, you will help shape the climate of the mathematics community, for good or bad.

Please choose to welcome and support your peers, and politely call out inappropriate behavior. Examples of inappropriate behavior include parading one's knowledge of advanced mathematics, dismissing others' questions as trivial, and any disparaging comments about other students' abilities, grades, appearance, course selection, likelihood to date, mathematical tastes, etc. If you don't feel comfortable doing this, then talk to me or some university official that you trust.

Finally, mathematics requires the courage to take risks. Many of us find it unnerving to share our mathematical ideas, especially those we are not sure are correct, with others. In order to develop our capacity for taking these risks, we must respect and support each other.²

Please also review our community agreement on Canvas.

Institutional policies: Please see cte.utah.edu/instructor-education/syllabus/institutional-policies.php or scan the QR code at right for information about university policies on the ADA act, Safety at the U, Addressing Sexual Misconduct, and Academic Misconduct.



²Thanks to Stephen DeBacker for help with some of the wording here.